		$F_f = 59.3$ N
7	**C**	Area under *F-t* graph gives the change in momentum of the body. From $t = 75$ ms to $t = 150$ ms, $\Delta p = mv - mu = ½\,(60 + 80)(100 - 75)(10^{-3}) + ½\,(80 + 40)(150 - 100)(10^{-3})$ $= 4.75$ $\therefore\ mv = 4.75 + mu = 4.75 + (0.200)(15) = 7.75$ $\therefore v = 7.75 / 0.200 = 38.8$ m s^{-1}
8	**A**	The resultant force acting on both blocks could be shown below.